

Yuexin (Oscar) Luo

Master Student, Mechanical Engineering (Control & Robotics), Carnegie Mellon University

Pittsburgh, PA, USA

☎ +1 540-750-5654 | ✉ oluo@andrew.cmu.edu

Summary

Master's student in Mechanical Engineering (Research) at Carnegie Mellon University, working on hip exoskeleton hardware and control as one inseparable problem: unstable or poorly-fitted hardware makes any controller impossible to evaluate fairly, and a controller that ignores human intent wastes even excellent hardware. Aiming to be equally strong at both, not specialized in one at the expense of the other.

Education

Carnegie Mellon University

M.S. in Mechanical Engineering – Research

- Focus: Control and Robotics | GPA: 3.96/4.0
- Advisor: Inseung Kang^[1]

Pittsburgh, PA, USA

Aug. 2025 – Expected May. 2027

Virginia Tech

B.S. in Mechanical Engineering

- Focus: Robotics and Mechatronics | GPA: 3.36/4.0

Blacksburg, VA, USA

Aug. 2019 – May 2023

Research Experience

Metamobility Lab, Carnegie Mellon University

Hip Exoskeleton Lead Designer

- Modified an off-the-shelf hip exoskeleton with custom motors and controllers (delayed-output feedback control, DOFC), achieving a metabolic benefit over no-exo baseline in single-subject pilot testing
- Replicated the off-the-shelf hip exoskeleton via 3D printing for the first design iteration, then reduced device weight from 4.2kg to 3kg over the next 3 months
- Diagnosed torque transmissibility loss ("motor rocking") at the hip joint through 26 build-test-analyze iterations; ruled out pelvic-interface contact area, then isolated excessive thigh-interface clearance as the cause
- Designed and fabricated a calf-surrounding thigh structure, eliminating rocking up to the motor's maximum output (17 N-m), up from onset at 12 N-m
- Identified garment and padding interface design as key to wearer comfort and fit through the same 26 iterations, cutting required belts and straps by half via the redesigned thigh structure

Pittsburgh, PA, USA

Sep. 2025 – Present

Metamobility Lab, Carnegie Mellon University

Contributing Researcher, Metabolic Reward Modeling

- Implemented five muscle-energy cost models as RL reward terms in a SCONE-based musculoskeletal control pipeline
- Found all five models produced similarly high-variance results, raising the question of developing a custom metabolic cost model

Pittsburgh, PA, USA

Jul. 2026 – Present

Metamobility Lab, Carnegie Mellon University

Independent Researcher, Gait Phase Estimation

- Implemented time-based gait estimation for exoskeleton control; found it failed to interrupt torque output on sudden stops, requiring a rolling-window adaptation strategy that still lagged behind speed changes
- Implemented a hybrid piecewise/unified phase-variable strategy; found asynchronous left/right leg phase-switching produced awkward transitions, and that torque commands computed from stale phase data persisted through sudden stops

Pittsburgh, PA, USA

Jun. 2026 – Present

Robotic Projects

DJI RoboMaster Competition, Virginia Tech

Blacksburg, VA, USA

Mechanical Engineer, "Hero" Sub-team

Sep. 2021 – May 2023

- Redesigned golf-ball launcher feed path from bottom-fed to rear-fed, cutting weight by 67% and assembly time by 40% while maintaining projectile capacity
- Built a mobile welded test chassis to field-test the launcher's target lock-on robustness under repeated handling
- Instructed 4 new team members on converting 3D CAD models into 2D CNC-manufacturable parts

Senior Capstone Design, MELD Automation Prep Line

Christiansburg, VA, USA

Mechanical Engineer

Aug. 2022 – May 2023

- Designed a pneumatic trap door to release cut bar stock into the next subsystem on cut detection, completing the transfer within 13% of the allotted cycle time
- Iterated corner-bracket geometry through several FEA-guided design cycles to meet the factor-of-safety requirement with a 2% margin
- Fabricated motor brackets by CNC mill and MIG welder as part of a cutting subsystem that can hold up to 3× desired mass

Intro Human-Robot Interaction Course Project, Best Pokémon Trainer

Blacksburg, VA, USA

Engineer

Mar. 2023 – May 2023

- Extracted Pokémon statistics to establish a database in Python, widening AI-generated Pokémon variation by around 400%
- Recovered reward weights with a standard deviation of 0.025 based on the Metropolis-Hastings method in Python
- Used inverse reinforcement learning (IRL) to train an AI with a 72% win rate against a simple AI and 45% against a human player

Work Experience

Soochow University Embedded System Lab

Suzhou, Jiangsu, China

Lab Assistant

Apr. 2024 – Mar. 2025

- Developed synchronous serial communication between two MCUs and a host computer using SPI as part of a graduate research project
- Developed a C# UI to translate data between characters, binary, decimal, and hex, improving debugging efficiency
- Prepared 68 PCBs by soldering electrical components for an undergraduate course

Hounen Solar America Inc.

Orangeburg, SC, USA

Equipment Engineer

Jul. 2023 – Dec. 2023

- Maintained and operated a solar cell stringer soldering machine, reducing unplanned downtime by 40%
- Analyzed and reduced assembly line blockage from twice a day to once per three days using FMEA
- Crafted standardized operating procedures for 4 machines, cutting employee training time by an average of 50%
- Instructed over 12 employees on machine operation along the assembly line

Skills

Embedded Systems

Arduino | STM 32 | Serial Communication (SPI, I2C, CAN)

Manufacturing

Soldering | Welding | Rapid Prototyping | 3-D printing | Machining | CNC | Laser Cutting

Software/Programming

Python | C++ | MATLAB | SolidWorks | Fusion 360 | Siemens NX | CATIA | OpenSim

Learning

ROS/ROS 2 | OpenCV | SCONE | Human-in-the-loop optimization

Certifications

SOLIDWORKS CAD Design Expert (CSWE), Dassault Systèmes, encompassing CSWP Weldments, Surfacing, and Sheet Metal specialties

Reference

^[1] Inseung Kang, Assistant Professor, Mechanical Engineering, Carnegie Mellon University, inseung@cmu.edu